

Artificial Intelligence usage declaration

In the context of this bachelor's thesis, Generative Artificial Intelligence tools were utilized in a targeted manner to support research, software engineering, and writing.

Detailed Task Allocation

Gemini Pro 3.1 — Complex Reasoning, Architecture, and Writing

The **Gemini Pro 3.1** model was utilized for phases requiring a comprehensive overview, synthesis skills, and the processing of extended contexts:

- **Literature review and state of the art:** Comparative analysis of standards, technical protocols, and scientific papers.
- **Architecture evaluation and comparative study:** Trade-off analysis (e.g., latency constraints, scalability, architectural choices).
- **Report structuring and outlining:** Logical organization of chapters and subsections to ensure a fluid and rigorous progression of the demonstration.
- **Translation and stylistic revision:** Global translation of the document and harmonization of the academic and technical linguistic register.

Gemini Flash 3.6 — Rapid Prototyping, Debugging, and Typst Diagrams

The **Gemini Flash 3.6** model was reserved for iterative tasks requiring immediate execution:

- **Code module prototyping:** Generation of function skeletons, configuration files, and unit test scripts.
- **Bug resolution and error analysis:** Rapid diagnosis of runtime errors, compilation exceptions, or network request malfunctions.
- **Generation of Typst schemas and diagrams:** Conversion of logical specifications into Typst source code (`#table`, `#grid`, vector components) that can be directly integrated into the report.

GitHub Copilot (VS Code Extension) — Real-Time Inline Assistance

Used daily via the VS Code extension as a contextual typing assistant:

- **Coding assistance:** Autocompletion of code lines, suggestions for repetitive structures, and generation of auxiliary functions.
- **Typst report writing assistance:** Dynamic completion during text, reference, and table/graphic tag entry within the IDE.